

MEASUREMENTS OF LIGHT STATISTICS WITH SPADs

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Abstract

This report documents the experimental characterization of Single Photon Avalanche Diodes (SPADs) and their application in classifying light sources via coincidence detection. Measurements utilizing pulsed lasers and pseudothermal sources extract photon statistics and validate Diffuse Correlation Spectroscopy (DCS). Throughout the document, hardware calibration, optical setups, and speckle decorrelation results are presented.

Introduction

Quantum optics classifies light sources based on their intrinsic photon emission statistics, which can be quantified using the second-order correlation function, $g^{(2)}(\tau)$. This function measures the time-separated probability of detecting two photons with a particular temporal delay τ . Different correlation functions of light sources allows us to study the statistical properties of light.

Phase-coherent light, is characterized by a constant macroscopic phase with independent photons governed by Poissonian statistics ($\sigma^2 = \mu$). The arrival of one photon provides no predictive information about the next, resulting in a flat correlation of $g^{(2)}(\tau) = 1$. In contrast, thermal light is formed by the superposition of countless independent waves with random, fluctuating phases, resulting in a Gaussian random field. The bosonic nature of photons in these chaotic fields leads to photon bunching—photons arriving in dense clumps followed by dark lulls. This broadens the distribution into super-Poissonian statistics ($\sigma^2 > \mu$) and yields a zero-delay correlation peak of $g^{(2)}(0) = 2$.

Because true thermal emission decorrelates on an ultrafast femtosecond timescale, capturing these chaotic phase fluctuations directly is limited by detector dead-time. However, by scattering a coherent continuous-wave laser

through a dynamic medium, such as a rotating diffuser, a “pseudothermal” light source is generated. This mechanically slows the random phase fluctuations into the millisecond regime, enabling the extraction of dynamic target velocities and scattering properties through Diffuse Correlation Spectroscopy (DCS).

By utilizing Time-Correlated Single Photon Counting (TCSPC) with Single Photon Avalanche Diode (SPAD) detectors and Time-to-Digital Converters (TDC), these temporal correlations can be accurately measured. When an incident photon strikes the active area of a SPAD detector, it triggers an avalanche of charge carriers, creating a macroscopic current pulse [Charbon, 2014, Antolovic et al., 2018, Hofbauer et al., 2018]. This signal is subsequently detected and converted into a highly precise digital timestamp by a TDC. Accumulating millions of these individual photo-detection events and referencing their arrival times constructs a $g^{(2)}(\tau)$ histogram.

This work uses SPADs and TDCs to measure the second-order correlation function and extract photon statistics and detection probabilities. These experiments serve as a tool to both distinguish the nature of light sources and perform a Diffuse Correlation Spectroscopy measurement.

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Equipment and Materials

Lasers and Light Sources

- Pulsed LASER: PicoQuant PDL Series
 - **PicoQuant Laser Diode Head**
 - **Wavelength:** 640 nm
 - **Repetition Rate (f_{rep}):** Tunable frequencies (operated between 1 MHz and 80 MHz).
- **Continuous Wave (CW) LASER 1**
 - **Thorlabs S1FC660 (Fiber-coupled)**
 - **Wavelength:** 660 nm
 - **Power:** Tunable from 0 mW to 19.8 mW
- **Continuous Wave (CW) LASER 2**
 - Thorlabs Benchtop Laser Diode Controller with mounted laser diode head.
 - **Wavelength:** 785 nm
 - **Power:** Tunable diode current (between 62.7 mA and 65.5 mA)
- **Pseudothermal Light Simulator**
 - Rotating glass diffuser (K10CR1/M). Motor Speed up to $20^\circ/s$.
 - Continuous Wave (CW) LASER 2

Timing Electronics

- Swabian Instruments Time Tagger
 - Model: Ultra, USB 3.0
 - Max Data Transfer Rate: 90 MTags/s
 - Max Input Frequency: 475 MHz
 - Dead Time: 2.1 ns
 - Base Jitter: 42 ps RMS (100 ps FWHM)
 - Trigger Level Range: -2.5 V to +2.5 V

Detectors and Optical Components

- **2 Pixel SPAD Module**
 - Model: NV02TPX
 - 64 bins of up to $5ns$ width
- **Linear SPAD Array Module**
 - **Model:** Novoviz Small Line Sensor
 - **Architecture:** 24-pixel linear SPAD array
- **Neutral Density Filters:** 0.6, 2.3

Layout and Hardware Calibration

The experimental layout used through the measurements is presented figure 1. The layout is designed to allow interchanging light sources incident on the SPAD. Optical elements can also be introduced between the source and the SPAD for attenuation, modulation and alignment purposes.

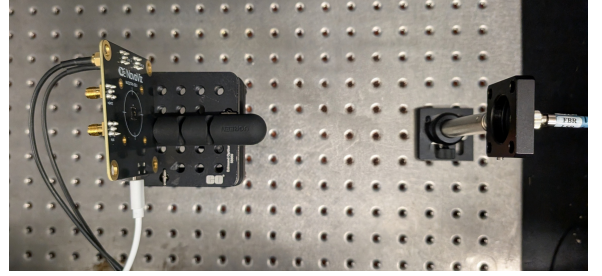


Figure 1: Experimental layout. Source is feed into the system with fiber optic cable that couples into free space with the SM1 FC2 and irradiates the SPAD.

0.1 SPAD Characterization

With the set up complete, a SPADs calibration is performed to set the operational baseline for the subsequent measurements. First, a dark count experiment where we run the `tt.CountRate` function. We integrate 205 measurements give dark counts of 9.74 *cps* and 10,301.49 *cps* for SPAD A and B respectively at this specific configuration. Next, we extract the dead time of the detector, and afterpulsing probabilities with the procedure listed below:

1. Connect the CW LASER source to the fiber optics line feeding the system.
2. Connect the SPAD detectors with SMA Cables to Swabian Time Tagger.
3. Initialize the autocorrelation measurement (`tt.Correlation`) using a single SPAD channel as both the "start" and "stop" trigger.
4. Wait for detection events to accumulate for 10 seconds.

5. Extract dead time and afterpulsing probabilities from data (See figure 2).
6. Repeat for 5 time to gather Statistics.

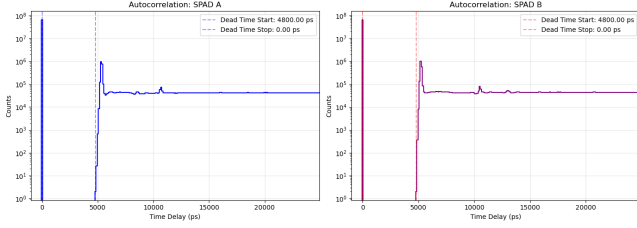


Figure 2: Autocorrelation histogram of the SPAD channels. Dead time is extracted from the temporal delay required for counts to recover from zero. Afterpulsing is extracted from taking the ratio of excess detections following the dead time to the total number of detections

The `tt.Correlation` time tags a series of detection events, calculates the time difference between all events (τ) and builds a histogram out of the number of detections that correspond to a particular time delay. Thus, from figure 2, we identify the total number of detections in bin $\tau = 0$ and the deadtime as the zero detection plateau. Afterpulsing is extracted as a ratio of excess detections that follow immediately after the deadtime to the total number of detections. The results are summarized in the table below:

Table 1: Summary of Detector Calibration Parameters

Parameter	SPAD 1	SPAD 2
DCR (cps)	9.74cps	10,301.49
Mean Dead Time (ns)	4.90 ± 0.12	4.85 ± 0.09
Max Theoretical Rate (MHz)	204.08	206.19
Afterpulsing Probability (%)	2.41	2.30

Poissonian Statistics Verification

0.2 Coherent Light

This section presents the experiment and measurements of the Poissonian and Super-Poissonian probability distributions. We followed the procedure below:

- Feed the CW LASER 1 into the optical setup in figure 1.
- Mount Filter in between the source and the SPAD.
- Define the temporal length of the windows integrating event counts: $100 \mu s$
- Connect SPAD to Swabian Time Tagger.
- Run the SWABIAN `tt.Counter` function and plot histogram [Swabian Instruments, 2026].

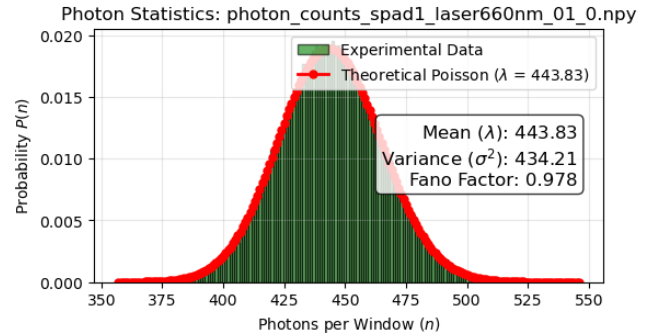


Figure 3: Photon counting statistics of the CW LASER at $1mW$.

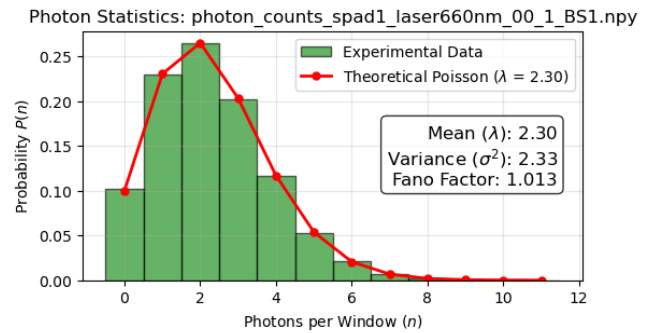


Figure 4: Photon counting statistics of the attenuated CW LASER.

As shown in figure 4, the coherent LASER source yields a poissonian distribution even

as we vary the power of the CW Laser. Furthermore, in figure 3, a polarizing beam splitter was introduced to measure the curve in low power regime; also resulting in a poissonian distribution. The fact that the variance matches the mean ($\sigma^2 \approx \mu$),¹, verifies the source as a coherent emitter.

0.3 Non Coherent Light

We proceed to switch the light source for a fluorescent light bulb and run the previous experiment with the CW LASER turned off. The resulting super-Poissonian distribution is shown in figure 5.

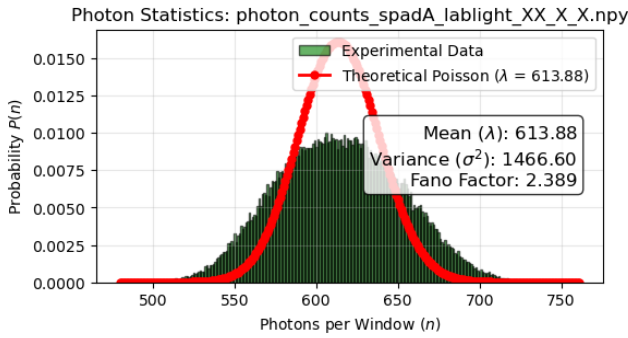


Figure 5: Photon counting statistics of the fluorescent lightbulb. The experimental data follows a super-Poisson distribution.

Coincidence Measurements

0.4 1st Measurement: HBT

First measurements measure second order coherence at $\tau = 0$ i.e $g^{(2)}(0)$. The procedure is given below.

1. Feed the Pulsed LASER into the optical setup in figure 1.
2. Select a LASER frequency: $1/T_{dead} \approx 2 \cdot 10^8 = 200MHz$.
3. Select Exposure Time.
4. Connect LASER Sync and SPAD channels into the time tagger.
5. Measure timestamps of detection events.
6. Compare Channel Ticks.

$$\Delta t_{coincidence} = \Delta t_B - \Delta t_A$$

$$\text{Coincidence if } \Delta t_{coincidence} \leq 2 \cdot \text{Jitter}$$

7. Calculate Probabilities

$$P_{A|B} = \frac{\#Events_{A|B}}{\#Pulses}$$

$$P_{AB} = \frac{\#Coincidences}{\#Pulses}$$

8. Calculate Correlation

$$g^{(2)}(0) = \frac{P_{AB}}{P_A \cdot P_B}$$

9. Check 3, 4 fold correlations $g^{(3)}, g^{(4)}, \dots$

This measurement configuration—cross-correlating the digital timestamps from two independent SPAD pixels to extract the intensity fluctuations—operates as a Hanbury Brown and Twiss (HBT) intensity interferometer [Hanbury Brown and Twiss, 1956]. By utilizing two distinct spatial channels, the setup measures coincident photon arrivals at zero temporal delay ($g^{(2)}(0)$).

¹Fano factor defined by $F = \frac{\lambda}{\sigma^2}$.

0.4.1 Pulsed LASER Measurements (05.06.2026)

Measurement of $g^2(\tau = 0)$ where taken with 10 ps resolution with a LASER attenuation adjusted to obtain 2% of total pulses detected by SPADs. Results are summarized in Figure 6.

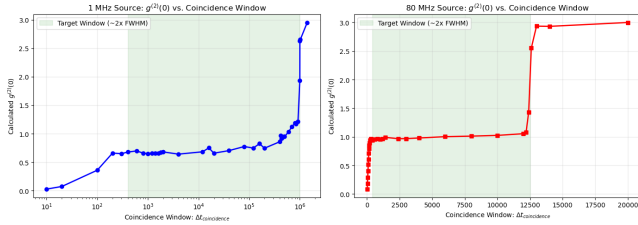


Figure 6: $g^2(0)$ measurement detailed in experiment 0.4.

These results were obtained in the laboratory session following the detailed steps below:

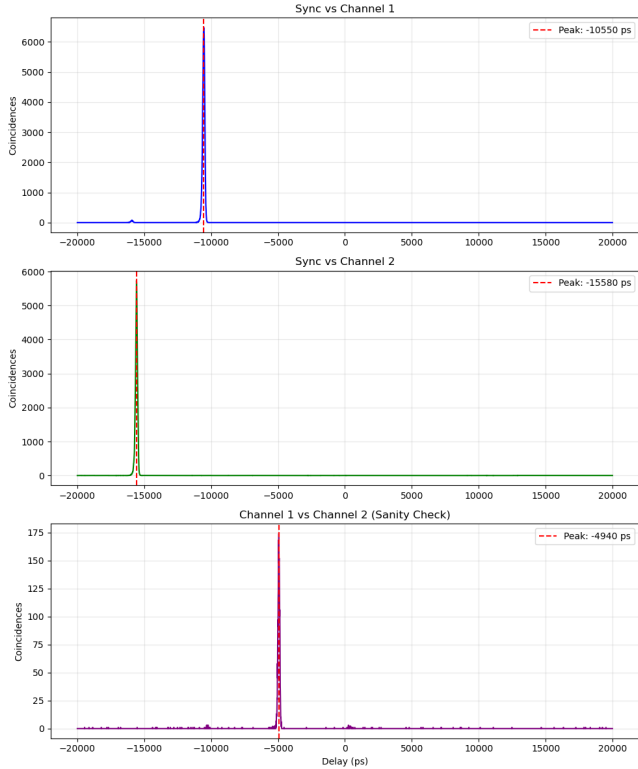


Figure 7: Differences in signal times. Calibration using `tt.Correlation` function

- **Hardware Setup:** Connect SPAD A, SPAD B, and LASER Sync to Swabian channels 1, 2, and 3 respectively. Align

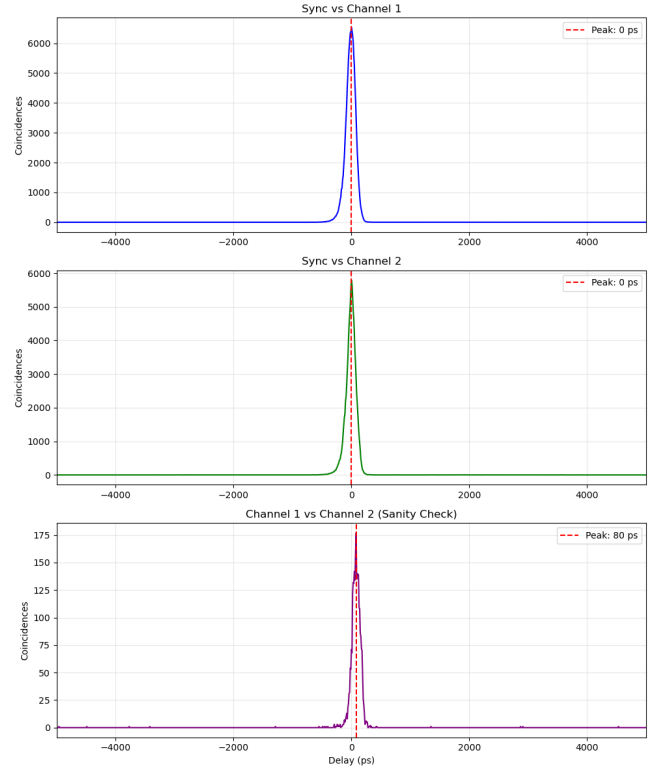


Figure 8: Calibrated signals to zero ps delay.

the LASER to the SPAD active areas, insert the ND0.6 filter, and isolate the system from external light.

- **1 MHz Configuration:** LASER configured to 1 MHz (*Source: Int2, Rep Frequency: 1*) at 36.8% power. Measured optical power is 3 μ W.
- **80 MHz Configuration:** LASER configured to 80 MHz (*Source: Int1, Rep Frequency: 1*) at 36.8% power. Measured optical power is 100 μ W.
- **Count Rate Calibration:** Averaged over 5×10^{11} ps using the `tt.CountRate` function. The baseline dark counts (LASER off) and active detections are summarized in Table 2.
- **Optical Path Delay Calibration:** Time differences between channels were extracted using the `tt.Correlation` function at 10 ps resolution. The peak delays are summarized in Table 3 (see Figure 7 and associated calibration plots).

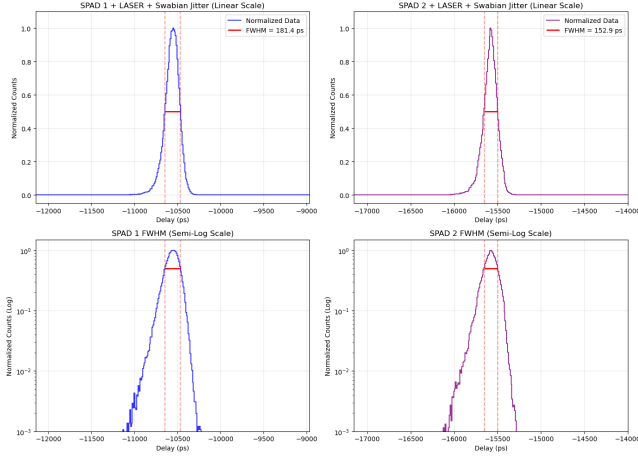


Figure 9: Optical Setup Jitter made up of Swabian, LASER & SPAD.

- **Delay Application:** Delays were fed back into the Swabian hardware via `tt.setInputDelay(SPAD, delay)` (see Figure 8).
- **$g^{(2)}(0)$ Measurement:** The second-order coherence measurement was executed per the procedure in Subsection 0.4. Results for both frequencies are referenced in Figure 6.

Table 2: Average Count Rates (cps) over 5×10^{11} ps

State	SPAD A cps	SPAD B cps	LASER cps
Dark Counts (LASER Off)	9.9	9,816.9	999,996.0
1 MHz (LASER On)	26,262.8	30,842.6	999,995.8
80 MHz (LASER On)	839,914.6	683,993.5	79,999,739.6

Table 3: Optical Path Delays via `tt.Correlation` (10 ps resolution)

Configuration	Sync → SPAD A	Sync → SPAD B	SPAD A → SPAD B
1 MHz Run	-10,550 ps	-15,580 ps	-4,940 ps
80 MHz Run	1,970 ps	-3,050 ps	-4,930 ps

0.4.2 Pulsed LASER Measurement (05.15.2026)

In Figure 6, we did not obtain the expected result $g^{(2)}(0)$ we were expecting. Probably this was due to the low operation power $3\mu W$. To test this hypothesis, we make a new measurement at 1 MHz with $33.5\mu W$. Results are shown in Figure 10.

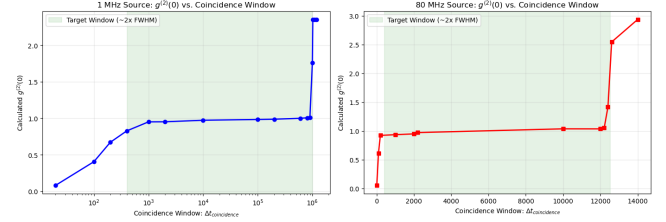


Figure 10: $g^{(2)}(0)$ measurement detailed in experiment 0.4.

- **Hardware Setup:** Connect SPAD A, SPAD B, and LASER Sync to Swabian channels 1, 2, and 3 respectively. Align the LASER to the active areas and isolate the system from external light.
- **1 MHz Configuration (High Power):** LASER configured to 1 MHz (*Source: Int2, Rep Frequency: 1*) at 100% power. Measured optical power is $33.63 \mu W$.
- **80 MHz Configuration:** LASER configured to 80 MHz (*Source: Int1, Rep Frequency: 1*) at 36.8% power. Measured optical power is $100 \mu W$. An ND0.6 filter is applied to throttle the active counts into the $\approx 1\%$ detection domain.
- **Count Rate Calibration:** Averages extracted using `tt.CountRate` over 5×10^{11} ps. The baseline dark counts and active detections for the multi-run 1 MHz measurements are summarized in Table 4.
- **Optical Path Delay Calibration:** Time differences extracted using `tt.Correlation` at 10 ps resolution. Peak delays for all runs are summarized in Table 5.
- **$g^{(2)}(0)$ Measurement:** Calculated delays were fed back into the Swabian hard-

were via `tt.setInputDelay()`. As expected for coherent Poissonian light, the measured $g^{(2)}(0)$ value successfully approaches 1 (see Figure 10).

Table 4: Average Count Rates (cps) over 5×10^{11} ps (1 MHz Source)

State	SPAD A cps	SPAD B cps	LASER cps
Dark Counts	9.3	9,542.9	999,996.0
1 MHz (LASER On)	665,354.0	584,594.0	999,996.0

Table 5: Optical Path Delays via `tt.Correlation` (10 ps resolution)

Configuration	Sync → SPAD A	Sync → SPAD B	SPAD A → SPAD B
1 MHz (Run 2)	-10,140 ps	-15,160 ps	-4,910 ps
1 MHz (Run 3)	-10,190 ps	-15,200 ps	-4,930 ps
80 MHz	1,970 ps	-3,050 ps	-4,930 ps

As a secondary result, we extract the system jitter from the Figures 7 and 8. As shown in Figure 9, jitter values for SPAD A and B are 181.4&152.1 respectively.

0.5 2nd Measurement: $g^{(2)}(\tau)$

Second measurements designed to extract the second order coherence function $g^{(2)}(\tau)$.

1. Feed the Pulsed LASER into the optical setup in figure 1.
2. Select a LASER frequency: $1/T_{dead} \approx 2 \cdot 10^8 = 200MHz$.
3. Build Histogram of $G^{(2)}(\tau)$
4. Normalize histogram by the average count of the time bins far from $\tau = 0$ to get $g^{(2)}(\tau)$.

For the Pulsed LASER we expect to see two behaviours depending on the programmed binwidth:

- If `binwidth` matches the period of the pulsed LASER we expect $g^{(2)}(0) = 1$.
- If `binwidth` is shorter than the period of the pulsed LASER, we expect a comb structure.

In contrast, for thermal sources, $1 < g^{(2)}(0) \leq 2$ and decays over coherence times. The width of this decay is a measure of the coherence time ².

0.5.1 Pulsed LASER Measurements (05.06.2026)

Results are summarized in Figures 15 and 17. Each figure comprises three subplots illustrating the data processing pipeline: the raw coincidence histogram $G^{(2)}(\tau)$ (grey), the correlation normalized via the Swabian API $g_{swabian}^{(2)}(\tau)$ (blue), and the manually normalized correlation $g_{python}^{(2)}(\tau)$ (red).

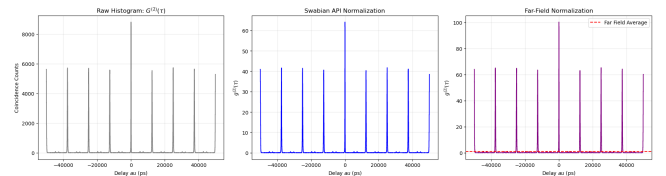


Figure 11: Autocorrelation function measurement for the LASER operating at 80MHz with its respective period of 12,500ps.

²Set by the diffuser speed

A similar result can be obtained for the 1MHz signal with its characteristic period shown in Figure 12.

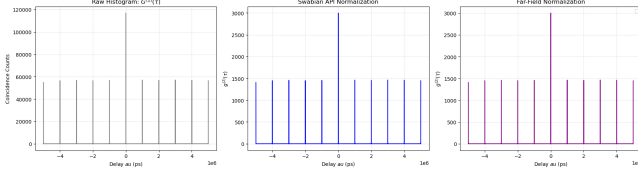


Figure 12: Autocorrelation function measurement for the LASER operating at 1MHz with its respective period of $1,000,000\text{ps}$.

Figures 11 and 12 are indicators that our setup is measuring the intended correlations. Note they are not properly normalized $g^{(2)}$ because the normalization is taking into account the secondary peaks. However, when we taylor the `binwidth` of the correlation function to match the LASER period, we observe a flat baseline (Figures 13 and 14).

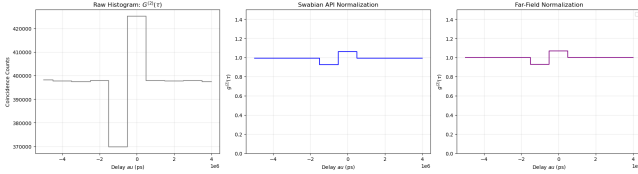


Figure 13: Autocorrelation function measurement.

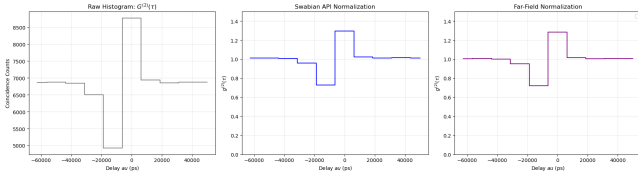


Figure 14: Autocorrelation function measurement.

Despite measuring the expected baseline in figures 13 and 14, they exhibit a severe quantization artifact: the digital Bin 0 overcounts, and consequently, Bin -1 significantly undercounts. This issue arises because the digital bin width is centered around 0 (i.e. $\tau_{bin0} = [-1/2f_{LASER}, +1/2f_{LASER}]$ rather than $\tau_{bin0} = [0, +1/f_{LASER}]$). To resolve this bin-edge quantization error, an artificial

hardware offset equal to exactly half the bin width was introduced via the input delays. This offset physically shifts the photon arrival times such that the coincidence peak lands dead-center inside Bin 0, safely away from the edges. The corrected measurement, successfully eliminating the binning artifacts, is presented in Figure 15.

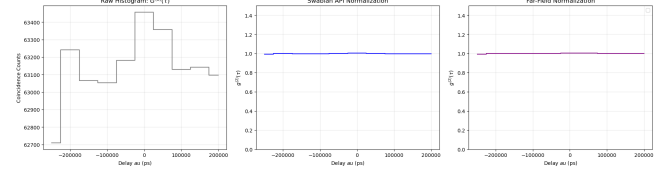


Figure 15: Autocorrelation function measurement.

While setting the correlation bin width equal to the LASER's repetition period yields a flat $g^{(2)}(\tau) = 1.0$ response, this must not be misinterpreted as continuous Poissonian emission. Rather, it is a temporal integration artifact where each macroscopic bin perfectly envelopes exactly one discrete pulse.

0.5.2 Thermal Source Measurements (05.12.2026)

Next we turn the attention to the thermal sources. We shine a thermal lamp into the SPAD and obtain its correlation function.

- Place 2.3 ND filter over the SPAD
- Turn on the Thermal Lamp or the Laboratory Light.
- Measure cps: 1,000,000
- Measure $G^{(2)}, g^{(2)}$
- Repeat for increasing cps.

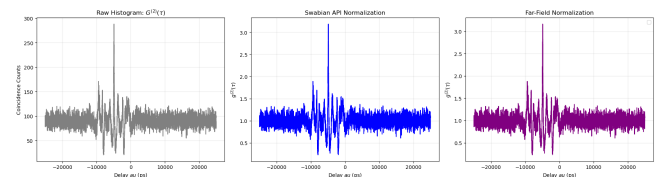


Figure 16: Autocorrelation function measurement for a thermal lamp.

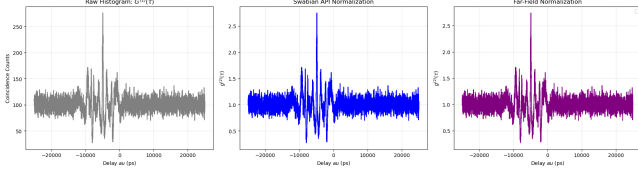


Figure 17: Autocorrelation function measurement for LabLight.

Note the oscillatory peak near $\tau = 0$ spanning approximately ± 10 ns in figures 17 and 16). This does not represent quantum thermal bunching, which theoretically decorrelates on a femtosecond timescale. Instead, we suggest that its a hardware artifact inherent to dense SPAD arrays, caused by a combination of optical crosstalk (hot-carrier luminescence between adjacent pixels). Outside of this nanosecond hardware interference window, the correlation function correctly rests at a flat baseline of $g^{(2)}(\tau) = 1.0$, confirming the system's inability to resolve ultrafast thermal statistics.

0.6 3rd Measurement: Diffuse Correlation Spectroscopy

As seen in the second measurement, true thermal light decorrelates in femtoseconds, which is too fast for the SPAD dead-time to measure. To solve this, we built a pseudo-thermal light source to measure photon bunching. We shined a 785 nm continuous-wave (CW) LASER through a rotating diffuser to create a moving speckle pattern. By changing the motor speed, we artificially slow down the light's intensity fluctuations to the millisecond regime, allowing our Time Tagger to capture it.

The measurement procedure was as follows:

1. A 785 nm CW LASER was aligned to hit the outer edge of a rotating diffuser.
2. The linear SPAD array was placed 0.5 cm behind the diffuser to detect the speckles.
3. To avoid hardware buffer overflow from high count rates, we lowered the LASER intensity to a safe kilocounts-per-second (kcps) level.

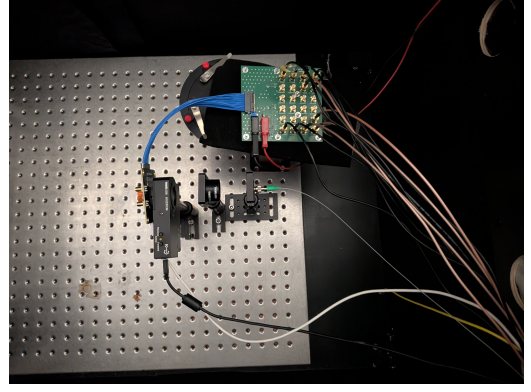


Figure 18: DCR setup measurement. CW LASER incident into diffuser with SPAD detector at the end (left).

4. To recover the Signal-to-Noise Ratio (SNR), we simultaneously measured the autocorrelation of 11 separate pixels to average them together.
5. The measurement was repeated for different diffuser angular velocities (5, 10, 15, and 20 deg/s).

Results

To process the data, each pixel's measurement was normalized by dividing it by its flat baseline. Figure 19 shows the 11 individual pixels (faded) and their combined average (black). Averaging the pixels gives a much smoother exponential decay curve, allowing us to accurately fit the data and extract the decorrelation time (τ_c).

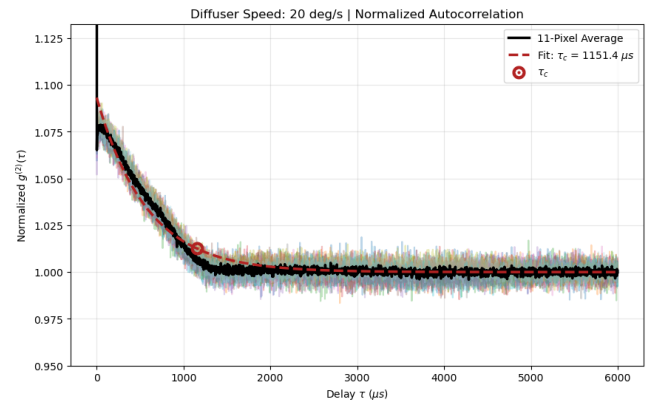


Figure 19: Normalized autocorrelation measurement for the rotating diffuser. The black line is the average of 11 pixels, which is fitted with an exponential curve (red dashed line) to extract τ_c .

Figure 20 demonstrates why using the 11-pixel array is so powerful. As we average more pixels (N), the measured decorrelation time remains stable, but the baseline noise drops follows the theoretical $1/\sqrt{N}$ rule.

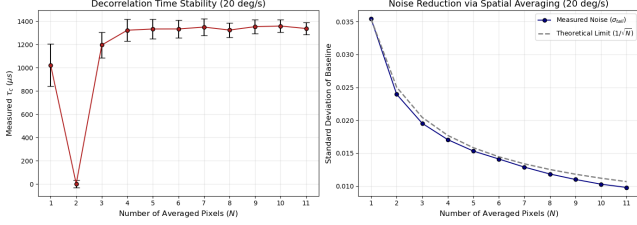


Figure 20: Effect of averaging multiple pixels. The measured τ_c remains stable (Left), while the noise drops with $1/\sqrt{N}$ (Right).

Finally, Figure 21 plots the extracted τ_c against the different motor speeds. As the diffuser spins faster, the speckles move past the detector faster, causing the decorrelation time to drop. This inverse relationship successfully validates our Diffuse Correlation Spectroscopy measurement.

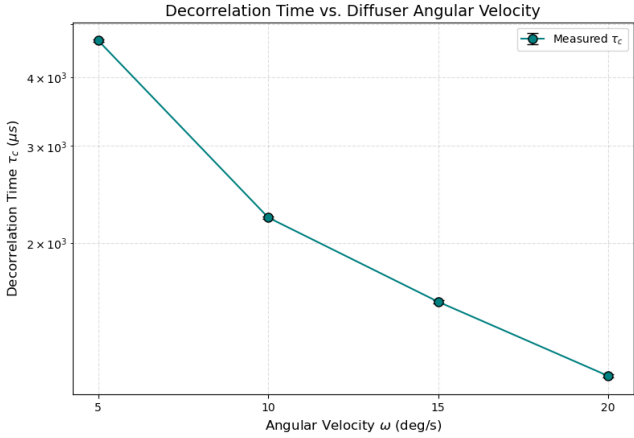


Figure 21: Decorrelation time (τ_c) measured at different diffuser speeds. As angular velocity increases, τ_c proportionally decreases.

Conclusion

This work demonstrated the use of Single Photon Avalanche Diodes (SPADs) and Time-Correlated Single Photon Counting (TCSPC) to measure the second-order coherence function, $g^{(2)}(\tau)$, of various light sources. By characterizing the detector hardware, the statistical differences between coherent lasers and super-Poissonian pseudothermal sources were discussed.

The measurement of true thermal emission (fluorescent lab light) highlighted the fundamental temporal limits of the detector architecture. The femtosecond decorrelation times of broad-bandwidth sources were shown to be entirely averaged out by the nanosecond dead-time and jitter of the SPADs, with any residual signal at zero-delay attributed to electronic and optical crosstalk between adjacent pixels rather than quantum photon bunching.

To bypass these ultrafast hardware limitations, a Diffuse Correlation Spectroscopy (DCS) setup was implemented. By scattering a continuous-wave laser through a rotating diffuser, chaotic phase fluctuations were artificially slowed into the millisecond regime, allowing the TCSPC hardware to accurately capture the exponential decorrelation of the dynamic speckle pattern. Furthermore, leveraging the linear SPAD array through multi-pixel spatial averaging proved highly effective in reducing baseline measurement noise without requiring extreme count rates. Ultimately, this methodology captured the dynamic photon statistics, accounted for detector artifacts, and successfully linked the extracted decorrelation times to the kinematic properties of the macroscopic target.

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